**ECSE 331**
**PSPICE Assignment No. 2**
**Simulation of semiconductor diode in PSPICE**[1]

## Objective

To get familiarized with simulation of semiconductor diode using PSPICE.

### Task 1. i-v characteristics of a diode

Construct the circuit shown in Fig. 1 in PSPICE. In the PSPICE parts library, you will find two components for D1N914: “D1N914/Design Cache” and “D1N914/Eval”. You can choose any one of the two. Sweep the value of the voltage source V1 over a reasonable range that allows you to find out the cut-off voltage of the diode. Plot the variation of curent through the diode with respect to the voltage across it. What is the cut-off voltage of the diode?

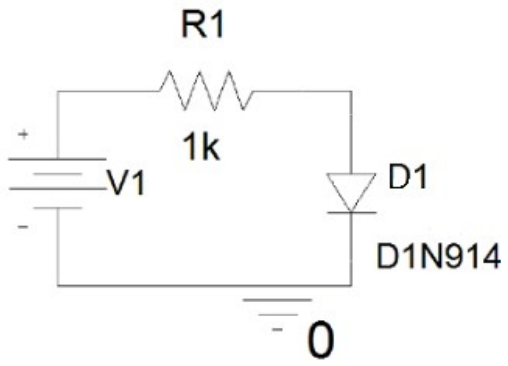



Fig. 1

### Task 2. Effect of temperature on diode characteristics

Edit the simulation profile as shown in Fig. 2 to simulate the circuit in Fig. 1 at three different temperatures: 50° C, 100° C and 150° C. Show the diode i-v characteristics plots at these temperatures and explain the graphs according to the theoretical description of diode behavior.

N.B: If you see the message "Only 1 expression allowed for the X variable" while trying to plot the y-axis results from different iterations, add @1 to the name of the x-axis variable. For example, if the original x-axis variable is V(D1:1), use V(D1:1)@1 in the x-axis in the plot.

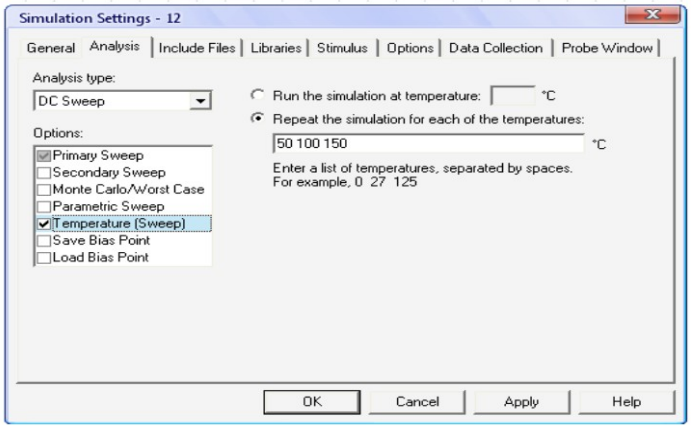



Fig. 2

1 The choice of PSPICE as the simulation software is not strict. The student may use any other simulation tool he/she is comfortable with.

**Task 3. Full Wave bridge rectifier**

Construct the circuit shown in Fig. 3 in PSPICE. For the transformer TX1, use the component XFRM_LINEAR/ ANALOG from the parts library. After placing TX1 on the schematic, double-click on it to open its property window and edit the properties as follows: COUPLING =1, L1_VALUE=100uH, L2_VALUE=1uH as shown in Fig. 4. V1 in a sinusoidal voltage source (VSIN). The diodes are the same at that used in task 1 and 2.

Obtain plots for (a) V1 versus time and (b) voltage across R1 versus time. Explain the waveform of the voltage across R1.

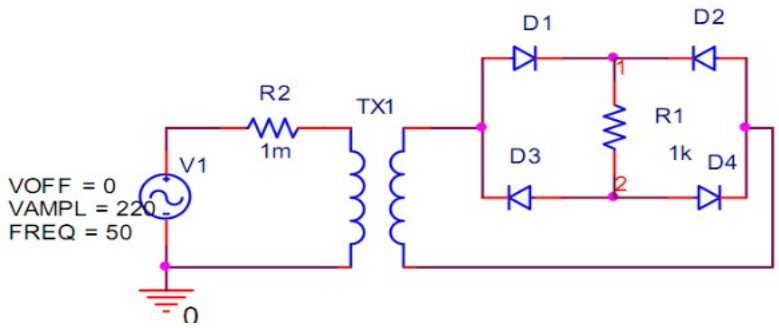



Fig. 3

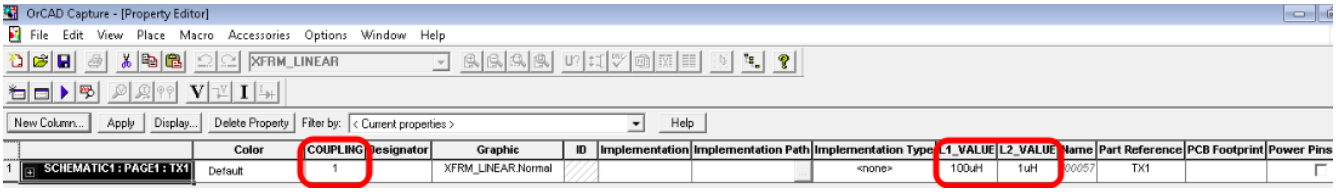



Fig. 4

**Task 4. Full Wave bridge rectifier with capacitor**

Add a capacitor of 10uF in parallel to R1 in the circuit shown in Fig. 3. Plot the voltage across R1 versus time. Compare this result with the result obtained in task 3 and explain the differences. Experiment with different values of the capacitor till you obatin a graph with negligible ripple. Show your results for at least 3 different values of the capacitor.

**Task 5. Clipper circuit**

For the circuit shown in Fig. 5, vin is a sinusoidal voltage source. Use the component “D1N4002/Design Cache” for the diodes. In this circuit, the values of V1 and V2 have to be chosen such that the output voltage V0 is as follows:

V0=vin for -5V<vin<9V
V0=9V for vin>9V
V0=-5V for vin<-5V

Find out what values of V1 and V2 should be used and justify your answer with relevant plots and theoretical discussion.

*N.B:* You should first find out the cut-off voltage for the diodes used in this circuit.

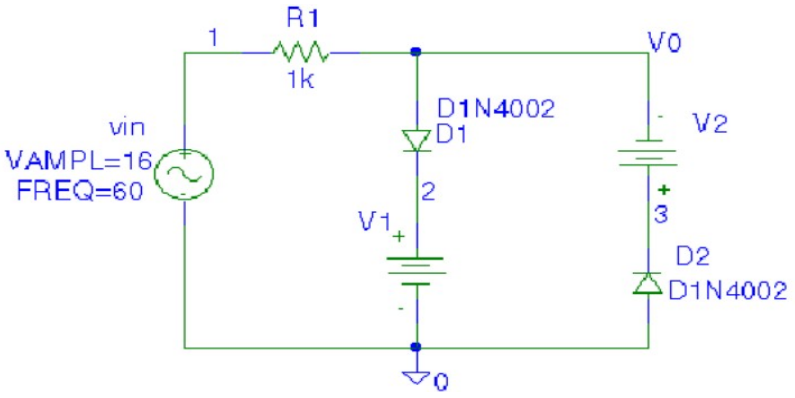



Fig. 5

**Task 6. Zener diode**

Construct the circuit shown in Fig. 6 in PSPICE. Sweep the value of the voltage source V1 over a reasonable range that allows the zener diode to go into forward conduction and reverse breakdown. Plot the variation of curent through the zener diode with respect to variation of voltage (Vout) across it. For what value of Vout does the zener breakdown occur? For what value of Vout does the zener diode start to conduct in the forward bias region?

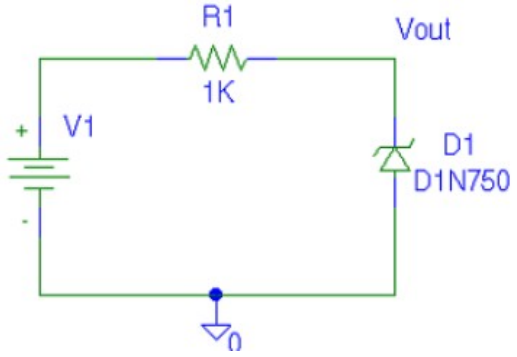



Fig. 6

**Write-Up Requirements**

A good laboratory report should contain a brief description of what the experiment was about, including circuit diagrams, and what you did, your data, your results, and anything else called for in the assignment, such as questions inserted in the laboratory. Answers to these questions require observations that need to be made at the time you do the experiment. The laboratory report should be written using the IEEE paper style consisting of a double-column single-space format, and should contain the following:

1. Title page - Title of the assignment/project, authors' name, and course name.
2. Abstract - Abstract of the assignment/project report.
3. Introduction
4. Main body of the assignment/project report including figures and explanations.
5. Conclusions
6. References (if needed)
7. Appendices (if needed)

For this assignment, the report should include screenshots of all the circuits built in PSPICE, screenshots of data reported from PSPICE, graphs from PSPICE and theoretical discussions wherever necessary to explain the results.

**END**